CSE 311:
Data Communication

Instructor:

Dr. Md. Monirul Islam

Signals and systems

Topics

- Definition, classification and properties of signals
- Examples of some useful signals
- Fourier Series of Periodic Signals

Properties of Fourier Transform

Time Differentiation Property

$$\frac{dg(t)}{dt} \Leftrightarrow j2\pi f G(f)$$

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$$\frac{d^n g(t)}{dt^n} \Leftrightarrow \left(j2\pi f\right)^n G(f)$$

Properties of Fourier Transform

Time Differentiation Property to find FT of complex signal

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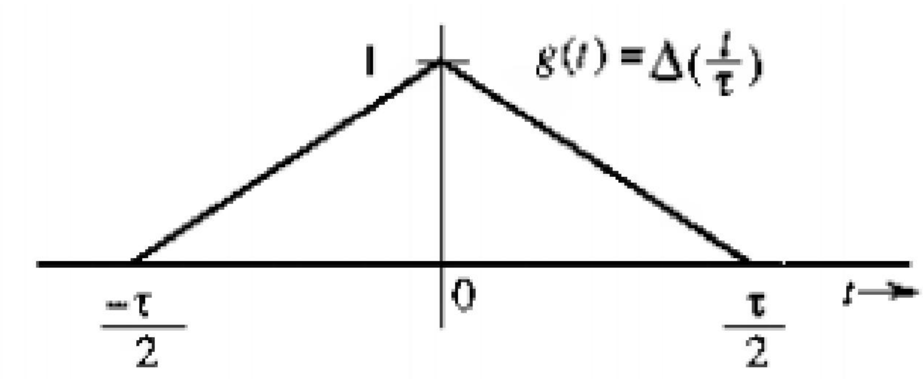

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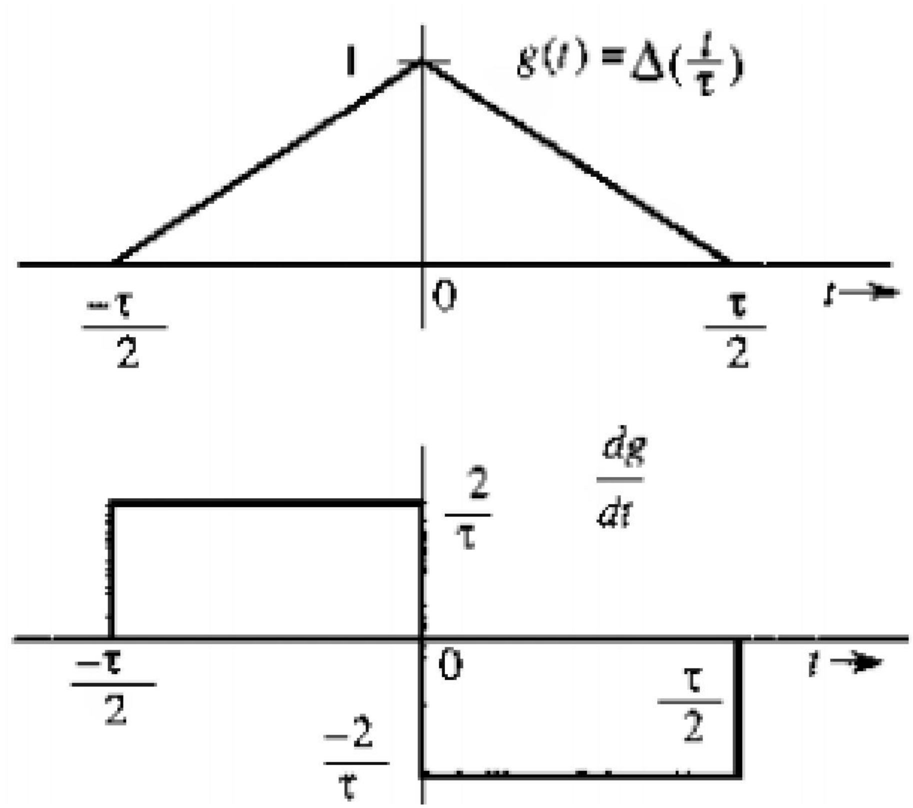

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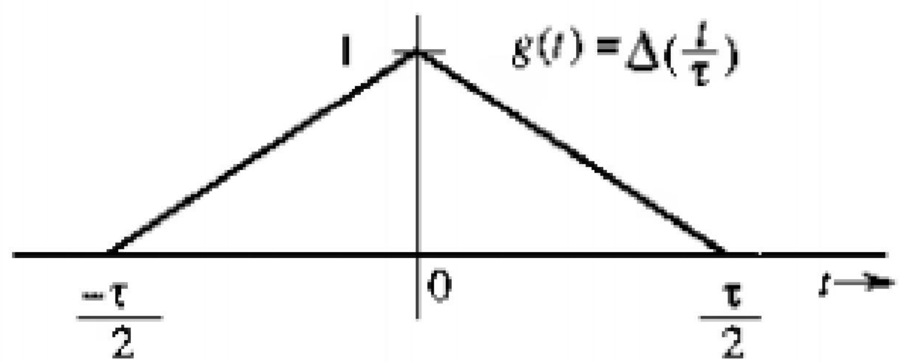

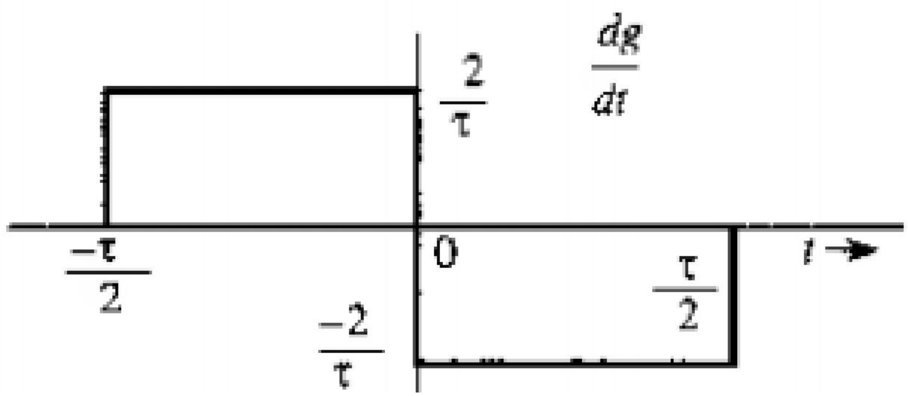

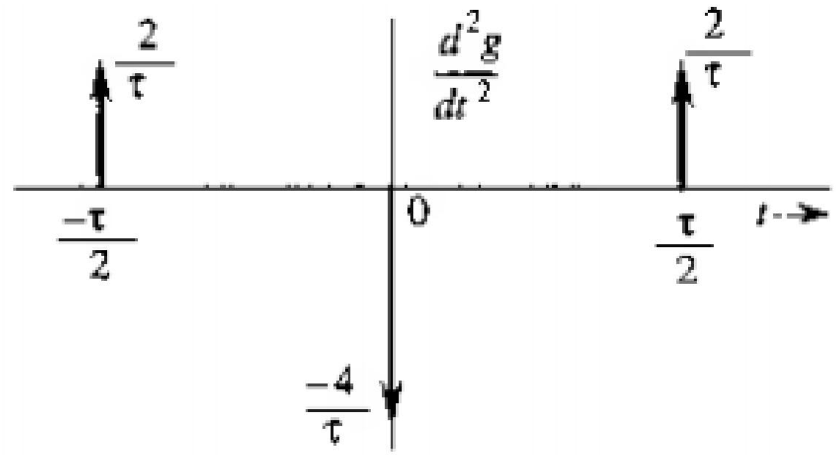

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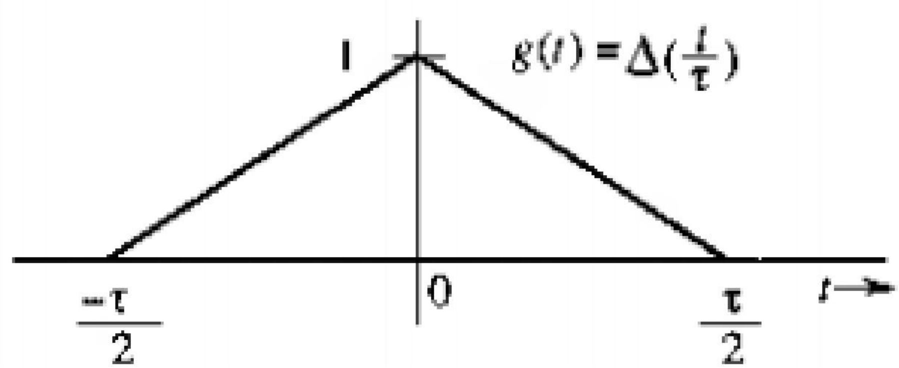

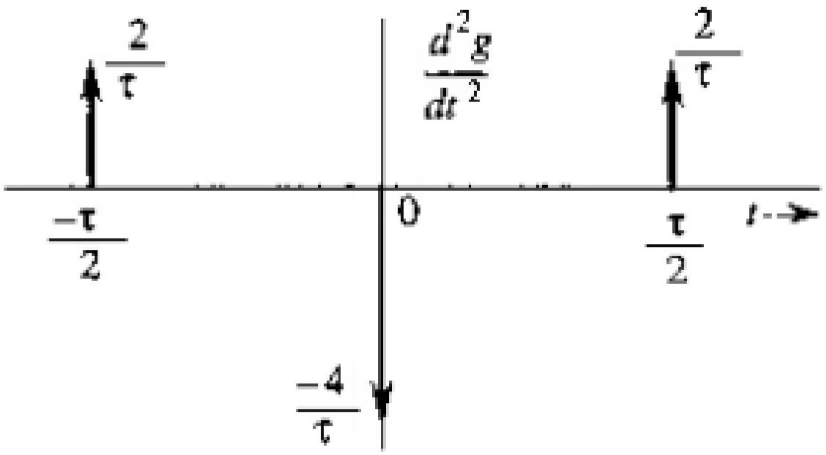

$$\text{Let, } \Im(\mathrm{g(t)}) = G(f)$$

Properties of Fourier Transform

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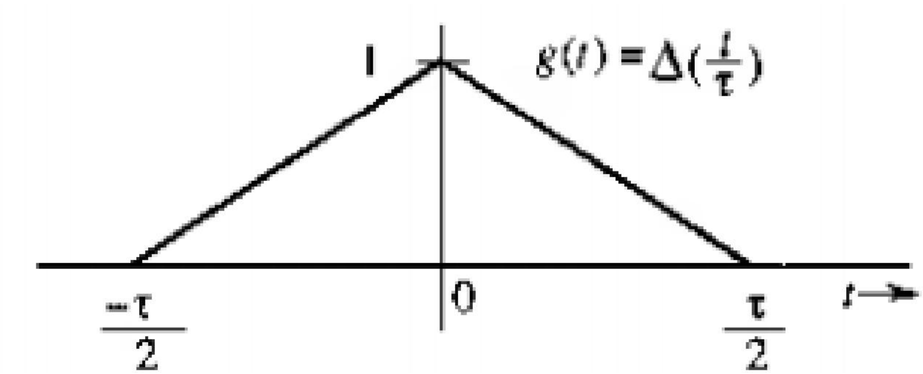

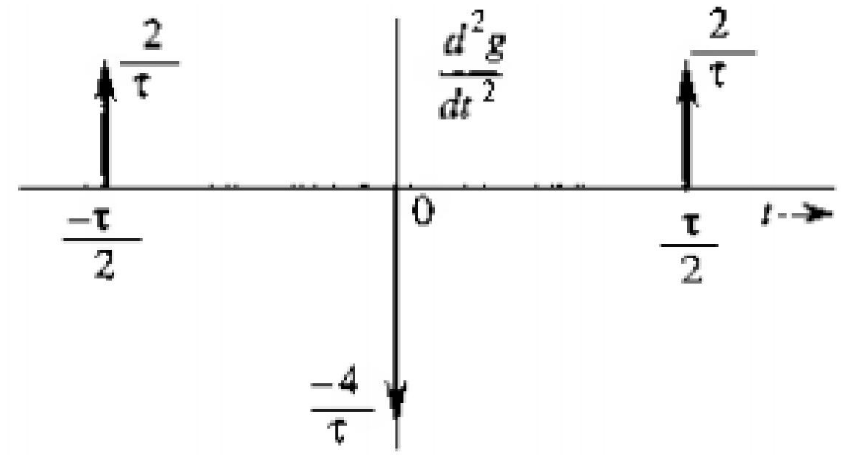

$$\frac{d^2 g(t)}{dt^2} = \frac{2}{\tau}\left[\delta\left(t+\frac{\tau}{2}\right) - 2\delta(t) + \delta\left(t-\frac{\tau}{2}\right)\right]$$

$$\frac{d^2 g}{dt^2} \Longleftrightarrow (j2\pi f)^2 G(f) = -(2\pi f)^2 G(f)$$

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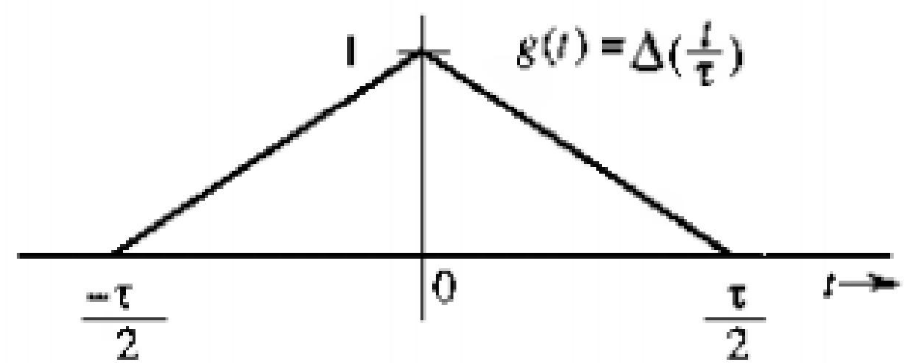

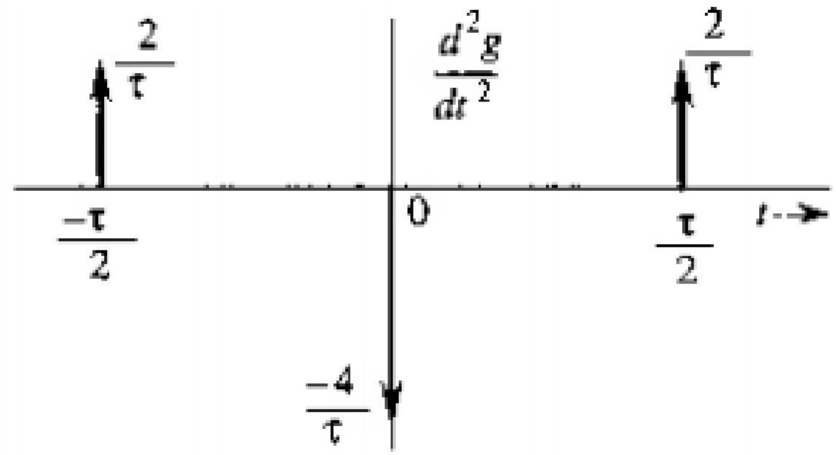

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$$\delta(t - t_0) \Longleftrightarrow e^{-j2\pi f t_0}$$

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$$\frac{d^2 g}{dt^2} \Longleftrightarrow (j2\pi f)^2 G(f) = -(2\pi f)^2 G(f)$$

$$(j2\pi f)^2 G(f) = \frac{2}{\tau}\left(e^{j\pi f \tau} - 2 + e^{-j\pi f \tau}\right)$$

$$G(f) = \frac{\tau}{2}\,\text{sinc}^2\left(\frac{\pi f \tau}{2}\right)$$

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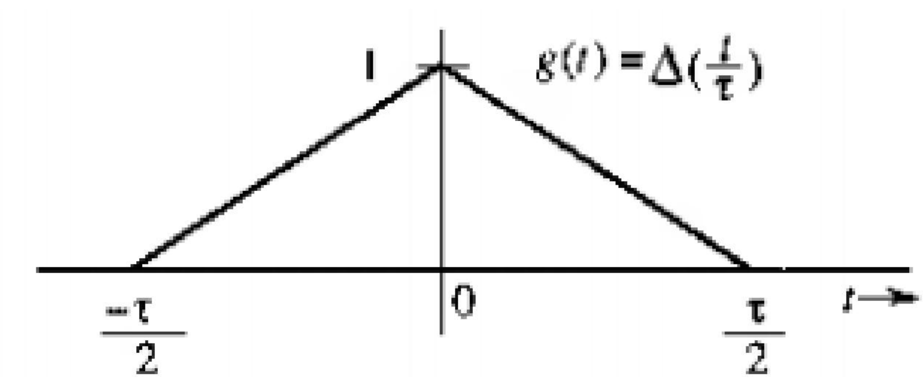
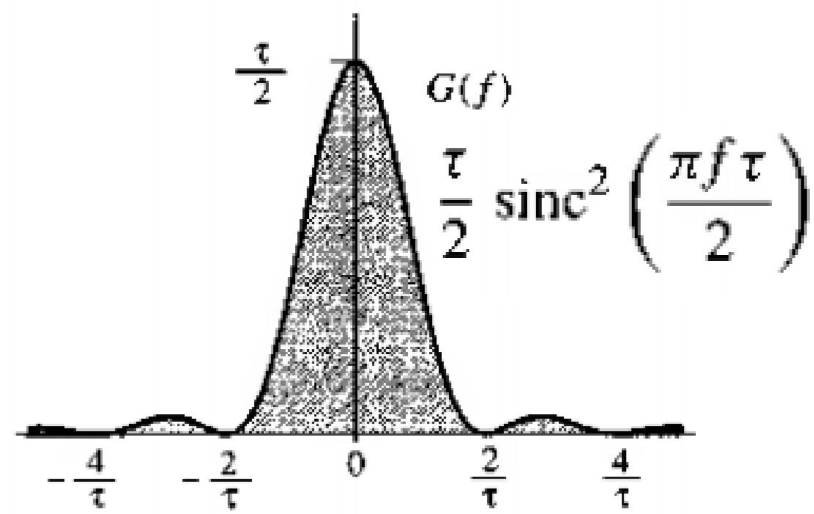
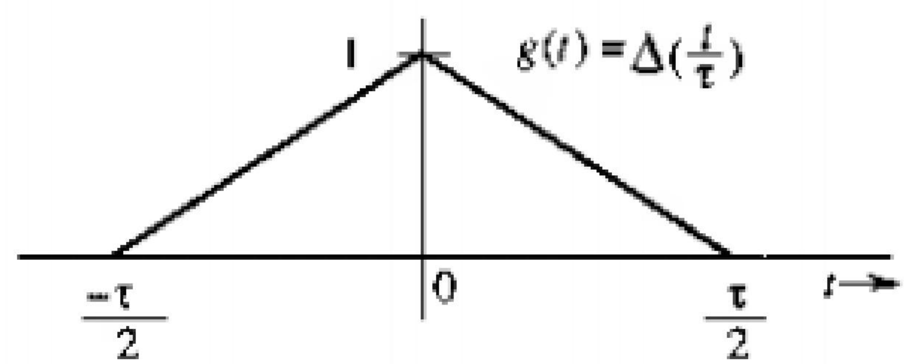
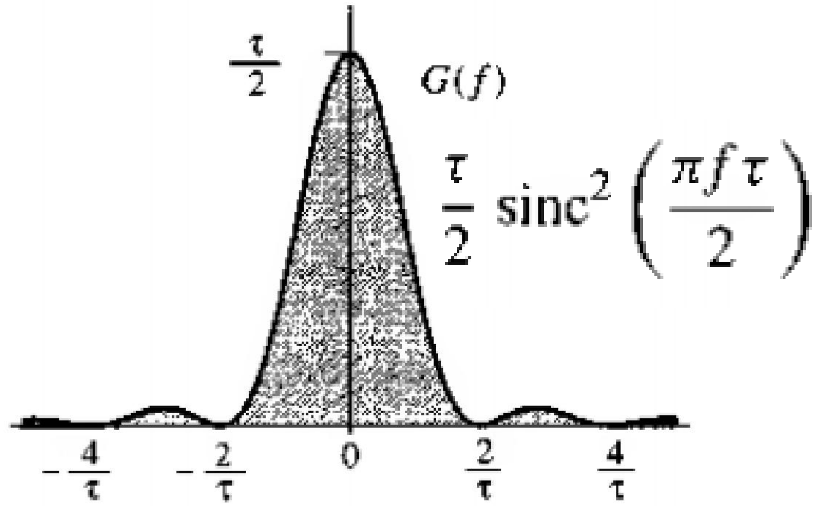




Time Domain and Frequency Domain Representation



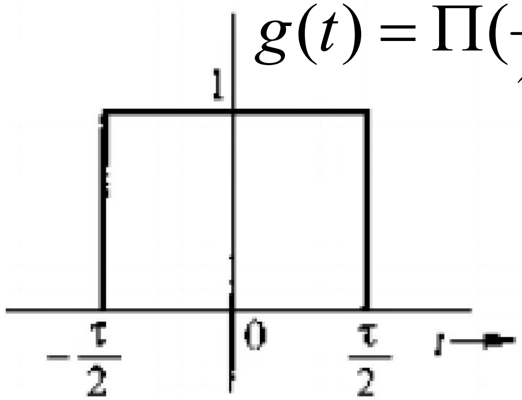
Limited in time



Indefinite spread in frequency

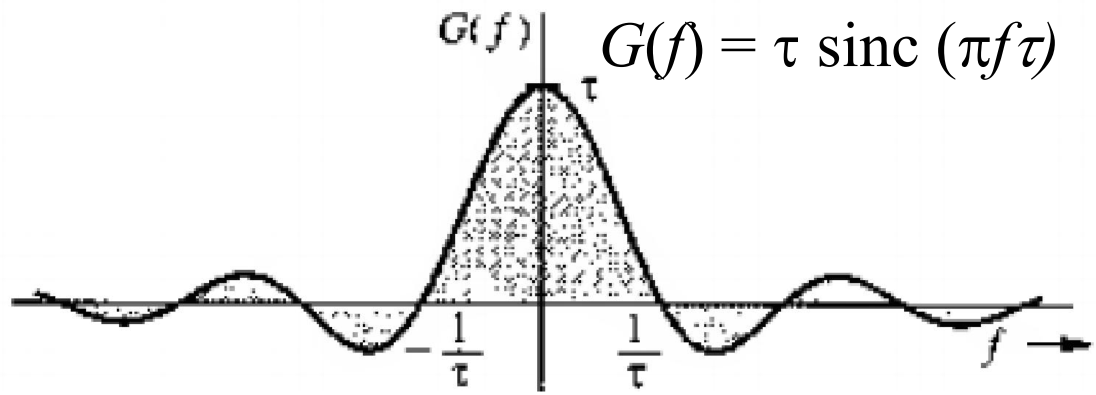
Time Domain and Frequency Domain Representation

$$g(t) = \Pi(\frac{t}{\tau})$$



$$\longleftrightarrow$$

$$G(f) = \tau \text{ sinc } (\pi f \tau)$$



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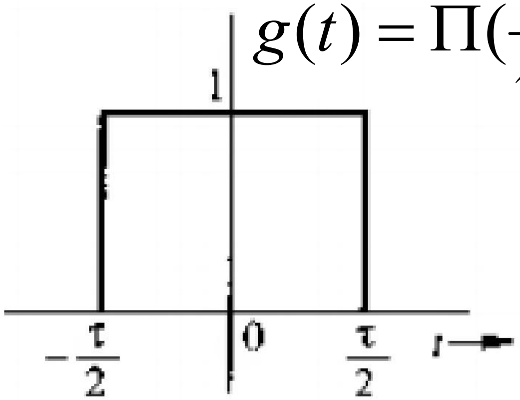

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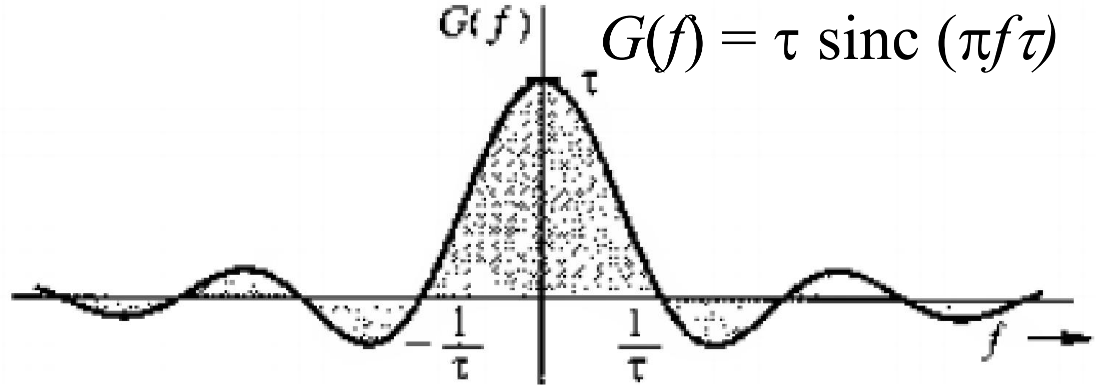
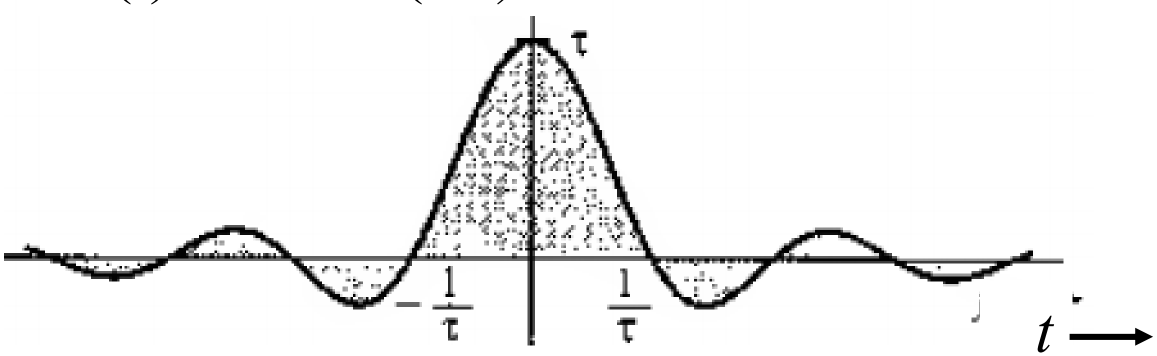


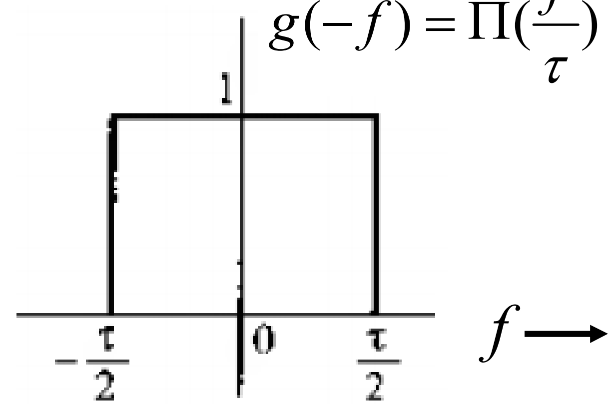
Indefinite spread in frequency

$$G(t) = \tau \text{ sinc } (\pi \tau t)$$



Indefinite spread in time

$$g(-f) = \Pi(\frac{f}{\tau})$$



Limited in frequency

Time Domain and Frequency Domain Representation

- A signal cannot be BOTH band limited and time limited
- Relation is inverse
- Relation changes in inverse manner
- Any change in time specification changes spectrum speciation inversely
- We can specify EITHER of them, NOT BOTH of them

Bandwidth of a Signal

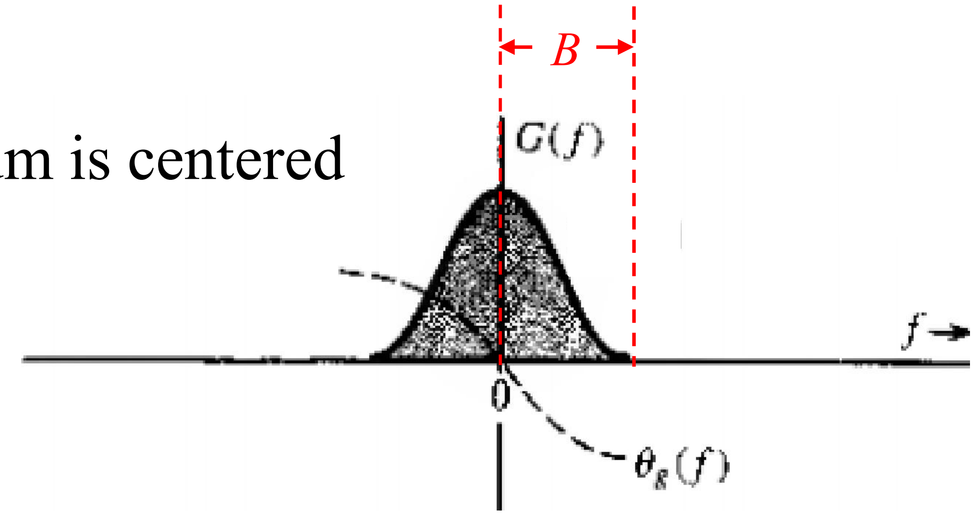
- Definition: extent of SIGNIFICANT spectral content of a signal for POSITIVE frequencies

- Alternate definitions exist

Bandwidth of a Signal

Lowpass signal:

Frequency spectrum is centered around $f = 0$



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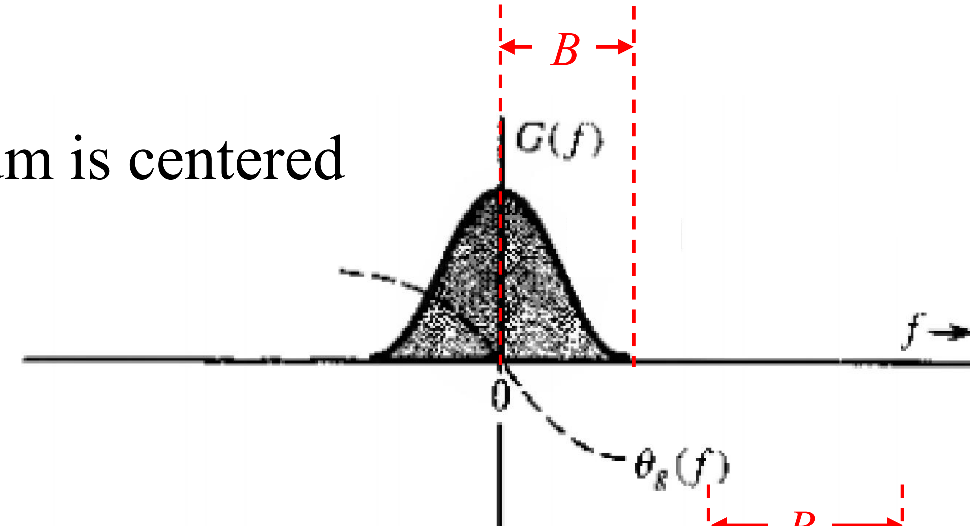
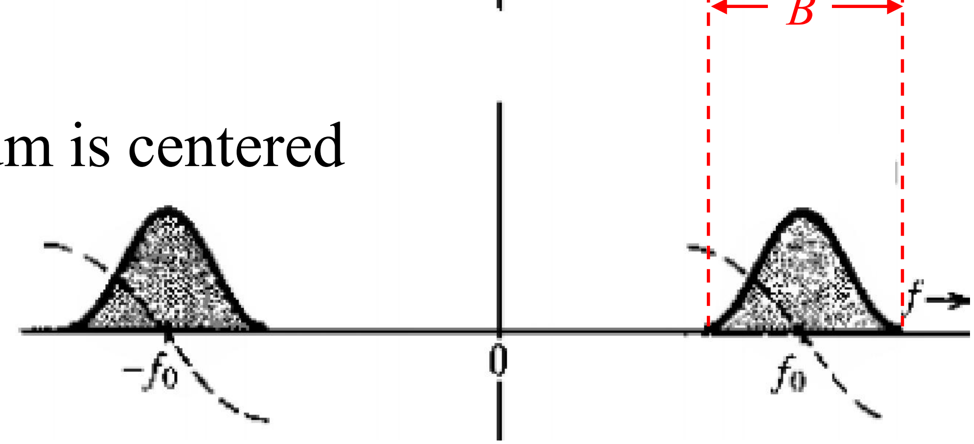


Bandpass signal:

Frequency spectrum is centered around $\pm f_0$



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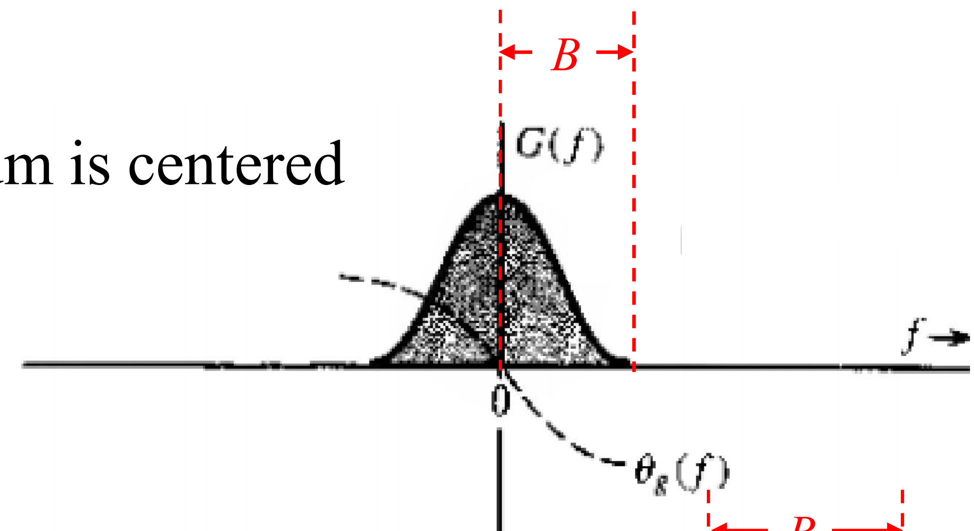
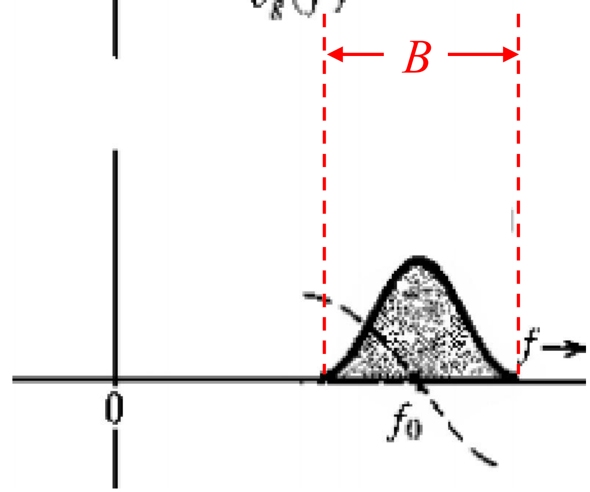
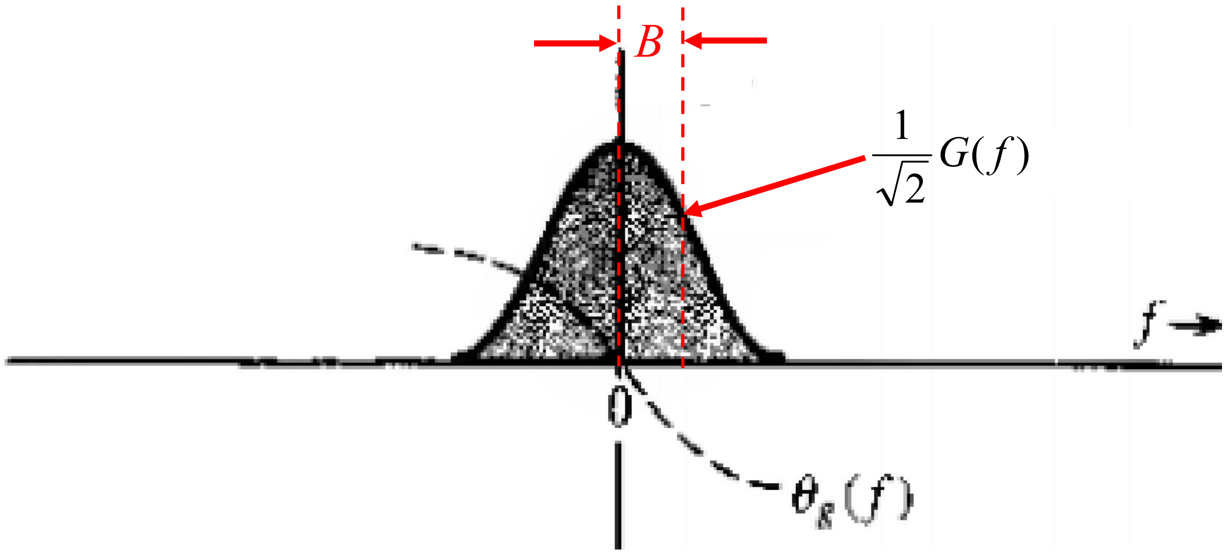


Shifting a low pass signal to f_0 location DOUBLES its bandwidth



3-dB Bandwidth of a Signal

Lowpass signal:
Separation betn peak and the position where it is $\frac{1}{\sqrt{2}}$ of its peak



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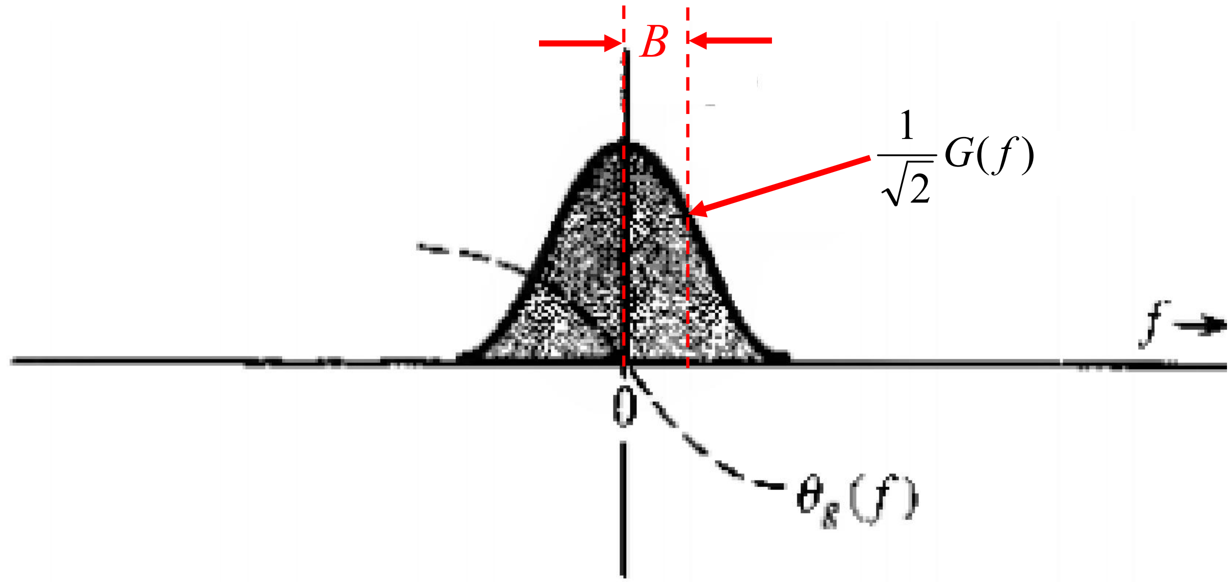
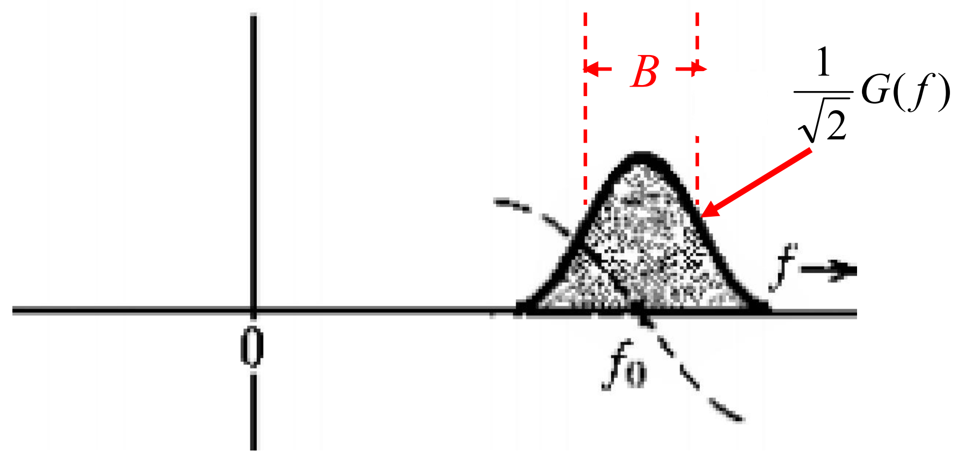


Bandpass signal:
Separation betn two positions where amplitudes are $\frac{1}{\sqrt{2}}$ of its peak



rms Bandwidth of a Signal

Root mean square bandwidth:

the square root of the second moment of a properly normalized form of the squared amplitude spectrum of the signal about a suitably chosen point

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k^{th}-moment of squared amplitude about origin $$= \int_{-\infty}^{\infty} f^k \left|G(f)\right|^2 df$$

rms Bandwidth of a Signal

Root mean square bandwidth:

the square root of the second moment of a properly normalized form of the squared amplitude spectrum of the signal about a suitably chosen point

2nd moment of squared amplitude about origin

$$m_2 = \int_{-\infty}^{\infty} f^2 |G(f)|^2 df$$

0-th moment of squared amplitude about origin

$$m_0 = \int_{-\infty}^{\infty} |G(f)|^2 df$$

rms Bandwidth of a Signal

Root mean square bandwidth:

the square root of the second moment of a properly normalized form of the squared amplitude spectrum of the signal about a suitably chosen point

2nd moment of normalized squared amplitude about origin

$$= \frac{\int_{-\infty}^{\infty} f^2 |G(f)|^2 \, df}{\int_{-\infty}^{\infty} |G(f)|^2 \, df} = \frac{m_2}{m_0}$$

rms Bandwidth of a Signal

Root mean square bandwidth:

the square root of the second moment of a properly normalized form of the squared amplitude spectrum of the signal about a suitably chosen point

$$rms \text{ bandwidth, } W_{rms} = \sqrt{\frac{\int_{-\infty}^{\infty} f^2 |G(f)|^2 df}{\int_{-\infty}^{\infty} |G(f)|^2 df}} = \sqrt{\frac{m_2}{m_0}}$$

Time-Bandwidth Product

$$\text{Duration} \times \text{bandwidth} = \text{constant}$$

- Relationship is true for every definition of bandwidth

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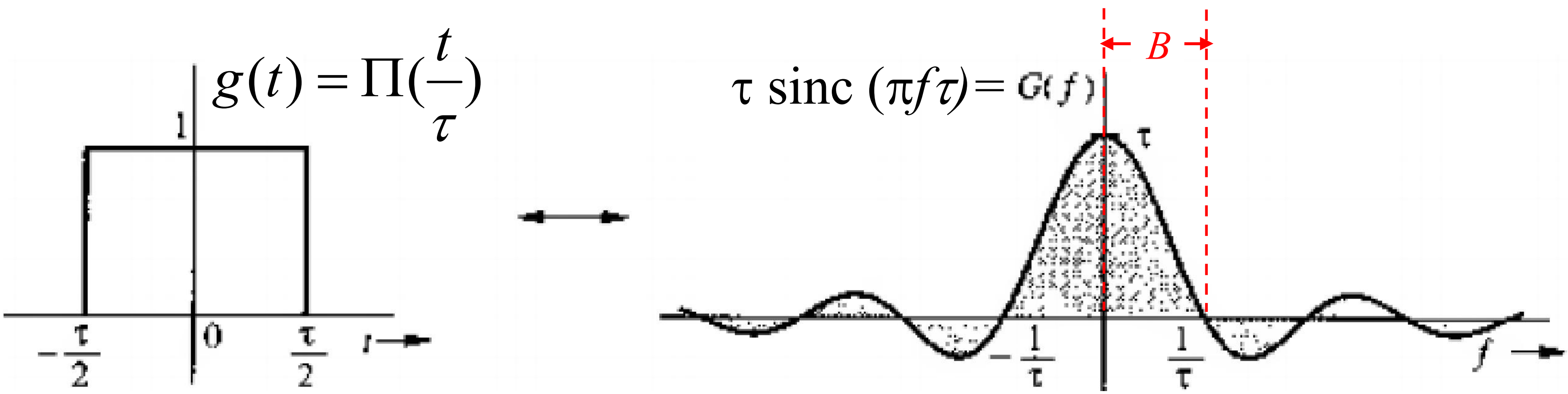
- Relationship is true for every definition of bandwidth
- Compression in time by a factor *a* expands the spectrum by the same factor and vice versa
    - No change in the constant

Time-Bandwidth Product

Duration × bandwidth = constant



Duration × bandwidth = τ× (1/ τ) =1 = constant

Time-Bandwidth Product

Duration × bandwidth = constant

$$T_{rms} = \sqrt{\frac{\int_{-\infty}^{\infty} t^2 |g(t)|^2 dt}{\int_{-\infty}^{\infty} |g(t)|^2 dt}}$$

$$W_{rms} = \sqrt{\frac{\int_{-\infty}^{\infty} f^2 |G(f)|^2 df}{\int_{-\infty}^{\infty} |G(f)|^2 df}}$$

Time-Bandwidth Product

Duration × bandwidth = constant

$$T_{rms} \times w_{rms} \geq \frac{1}{4\pi}$$

Time-Bandwidth Product

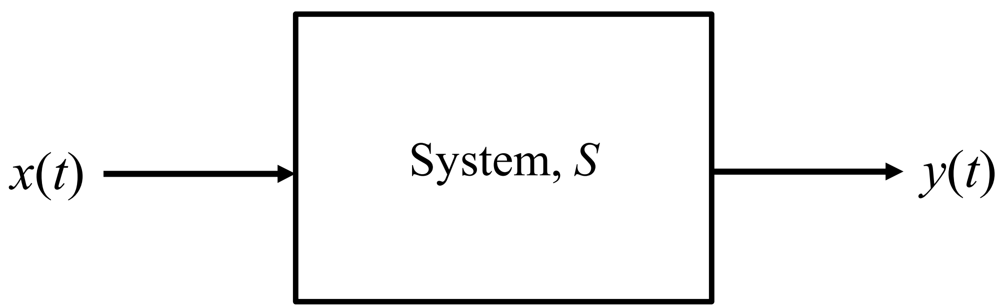
Duration × bandwidth = constant

$$T_{rms} \times w_{rms} \geq \frac{1}{4\pi}$$

- Equality holds for Gaussian pulse

Signal Transmission through Linear Time Invariant System

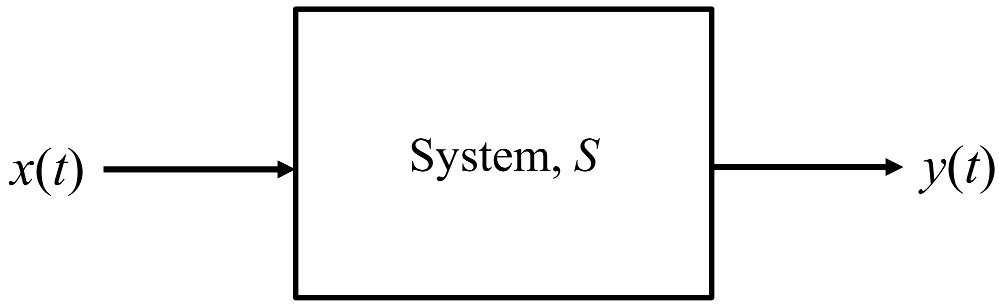
System



$$y(t) = S(x(t))$$

Signal Transmission through Linear Time Invariant System

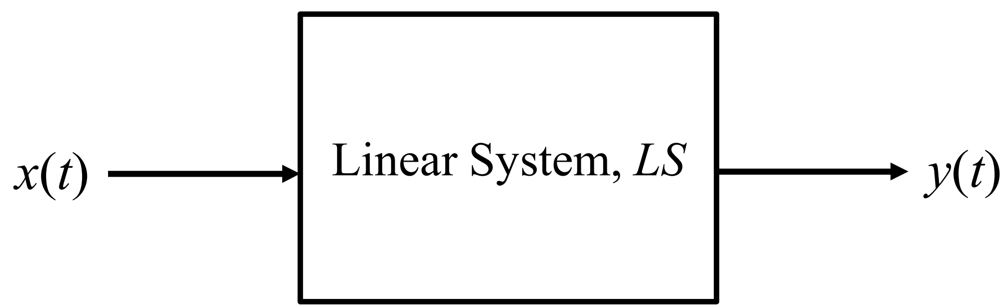
System



$$
\begin{aligned}
y(t) &= S(x(t)) \\
y_1(t) &= S(x_1(t)) \\
y_2(t) &= S(x_2(t))
\end{aligned}
$$

Signal Transmission through Linear Time Invariant System

Linear System

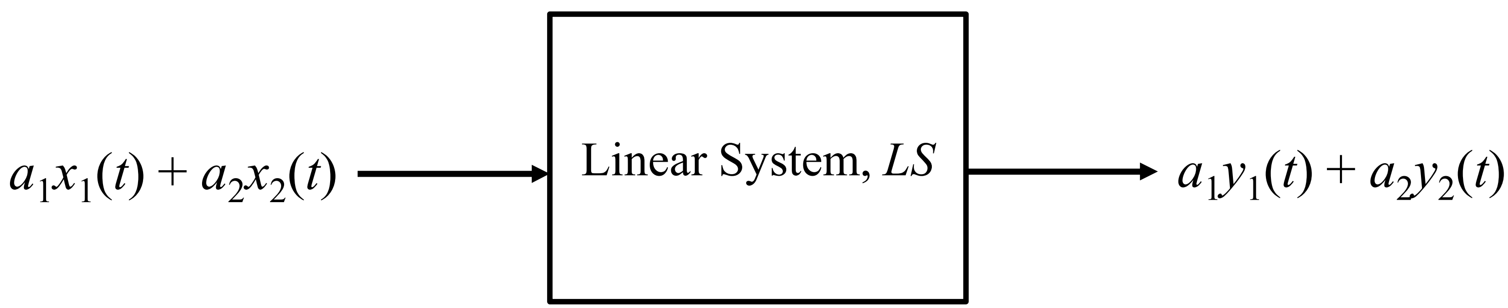


$$y(t) = LS(x(t))$$

$$y_1(t) = LS(x_1(t)) \qquad\qquad y_2(t) = LS(x_2(t))$$

Signal Transmission through Linear Time Invariant System

Linear System



$$y(t) = LS(x(t))$$

$$y_1(t) = LS(x_1(t)) \qquad y_2(t) = LS(x_2(t))$$

$$a_1y_1(t) + a_2y_2(t) = LS(a_1x_1(t) + a_2x_2(t))$$

Signal Transmission through Linear Time Invariant System

Linear System

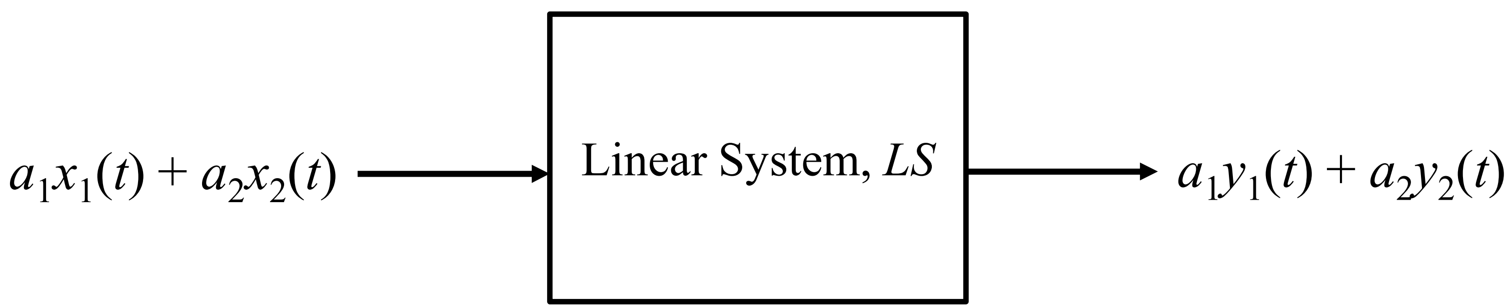
$$a_1x_1(t) + a_2x_2(t) \longrightarrow \boxed{\text{Linear System, } LS} \longrightarrow a_1y_1(t) + a_2y_2(t)$$

Principle of Superposition

the response of a linear system to a number of excitations applied simultaneously is equal to the sum of the responses of the system when each excitation is applied individually.

Signal Transmission through Linear Time Invariant System

Linear System



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Examples

Filters and Communication Channels

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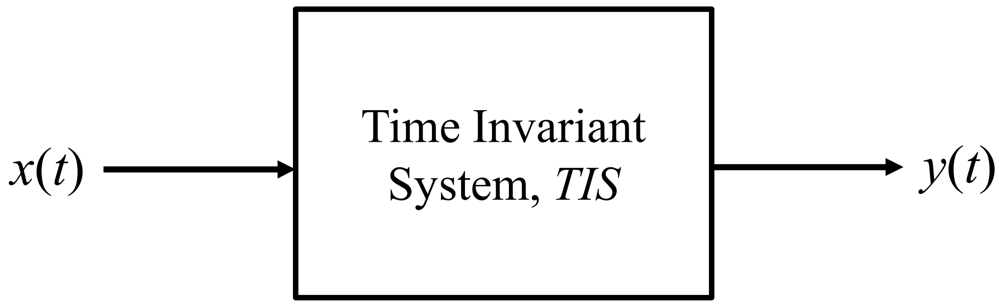
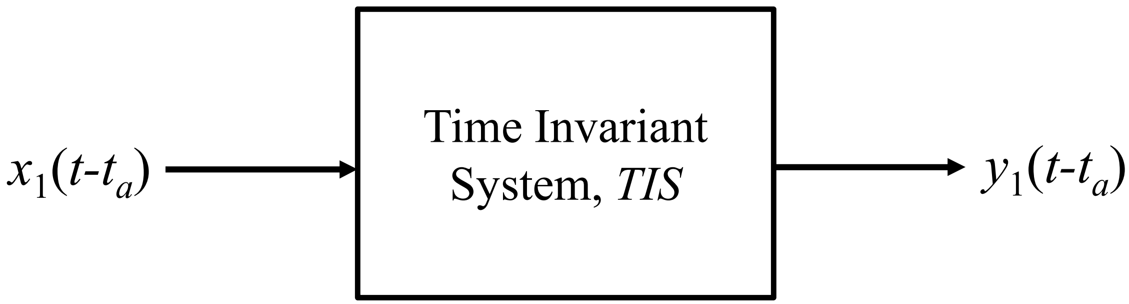
Time Invariant System



$$y(t) = TIS(x(t))$$

Signal Transmission through Linear Time Invariant System

Time Invariant System

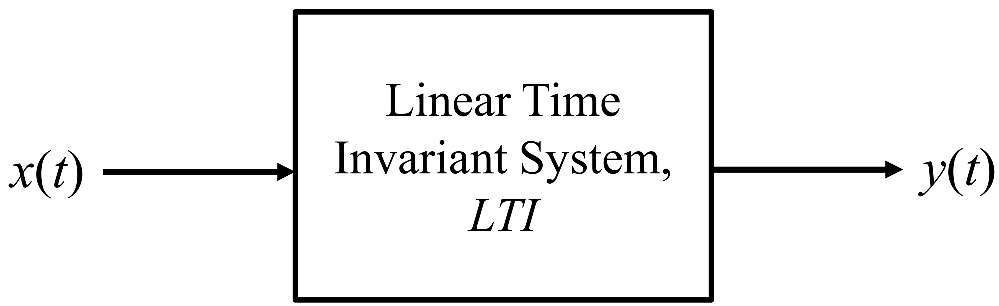


$$y(t) = TIS(x(t))$$

$$y_1(t\text{-}t_a) = TIS(x_1(t\text{-}t_a)) \qquad y_2(t\text{-}t_b) = TIS(x_2(t\text{-}t_b))$$

Signal Transmission through Linear Time Invariant System

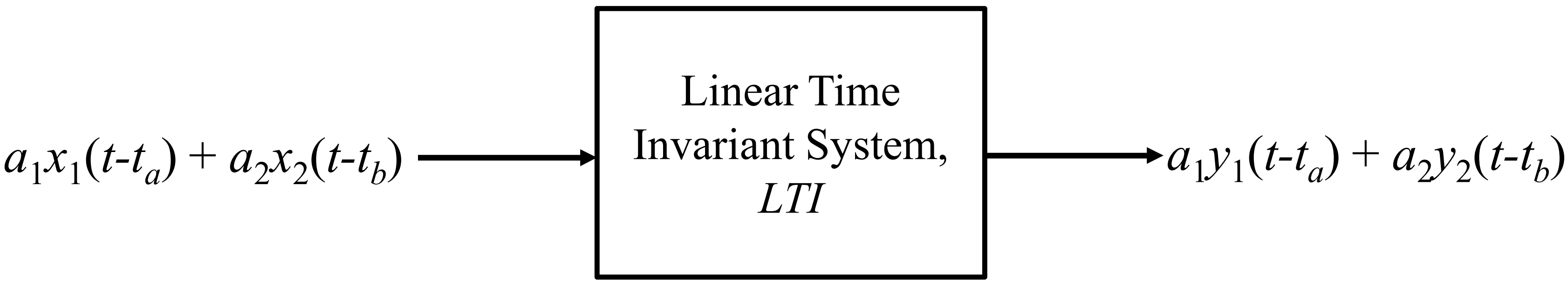
Linear Time Invariant System



$$y(t) = LTI(x(t))$$

Signal Transmission through Linear Time Invariant System

Linear Time Invariant System

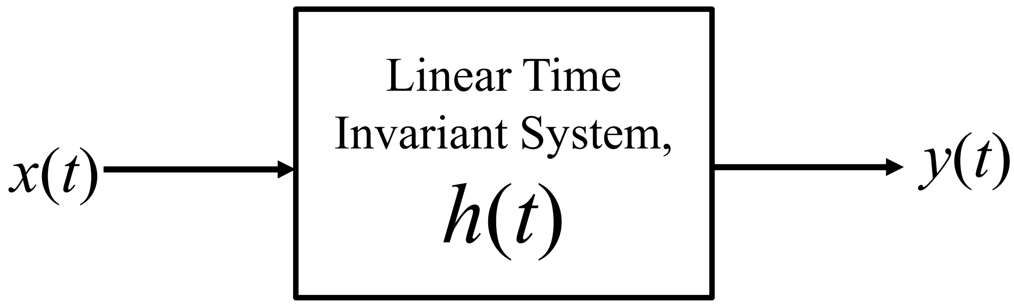


$$y(t) \; = LTI(x(t))$$

$$a_1y_1(t\text{-}t_a) + a_2y_2(t\text{-}t_b) \; = LTI(a_1x_1(t\text{-}t_a) + a_2x_2(t\text{-}t_b))$$

Signal Transmission through Linear Time Invariant System

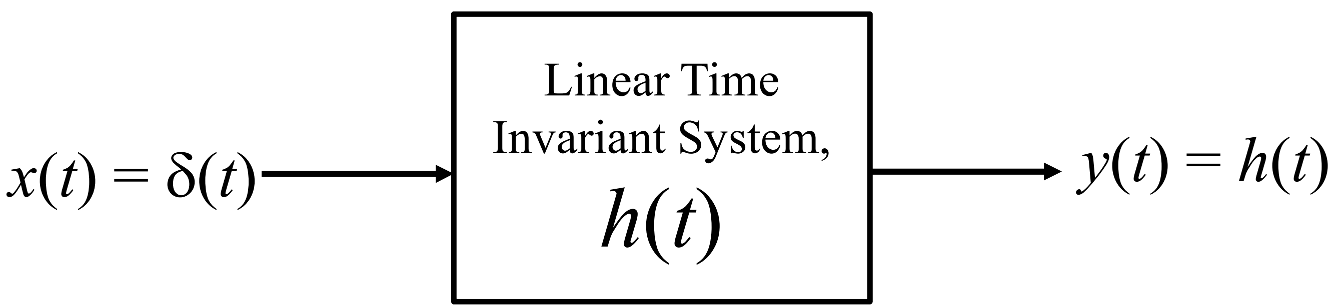
Linear Time Invariant System



LTI system is described by impulse response, $h(t)$

Signal Transmission through Linear Time Invariant System

Linear Time Invariant System



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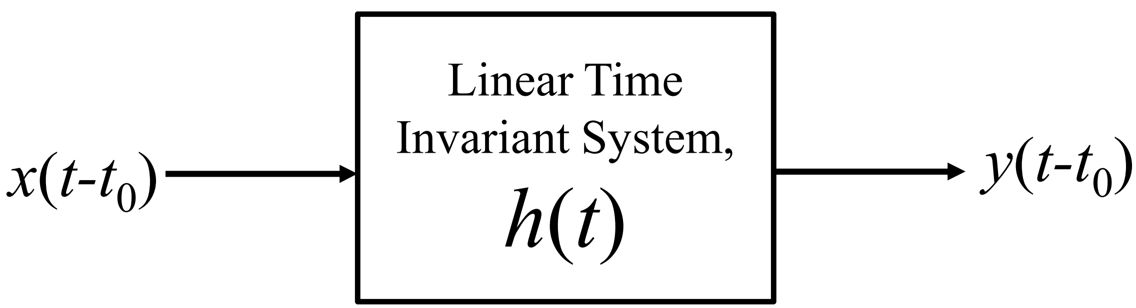

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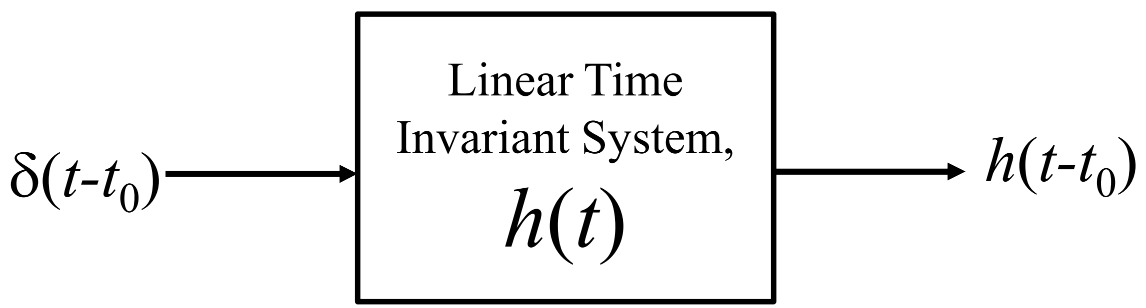

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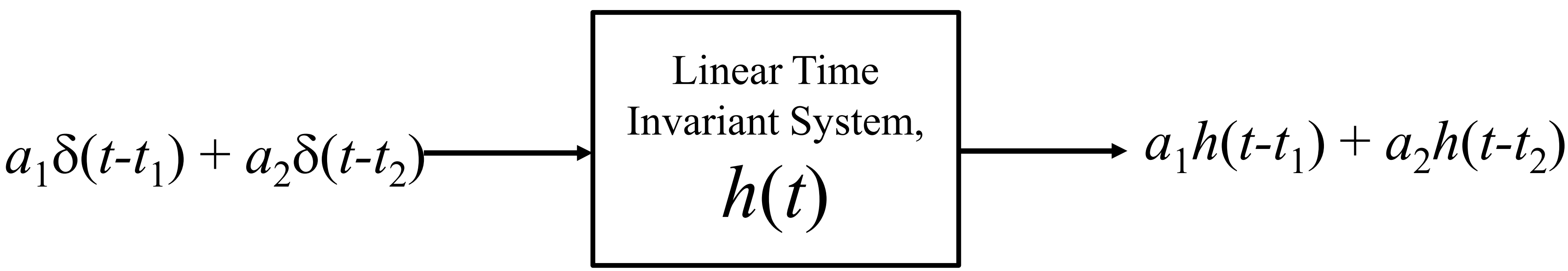
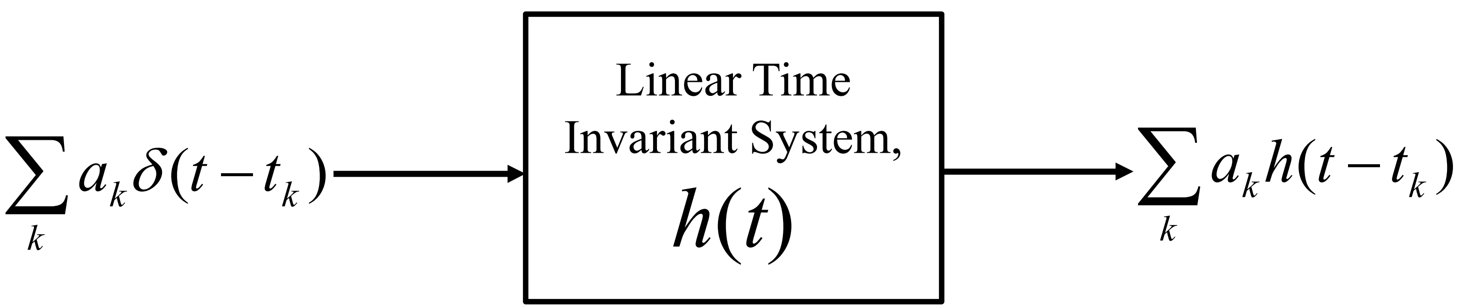


LTI system is described by impulse response, $h(t)$

Signal Transmission through Linear Time Invariant System

Linear Time Invariant System



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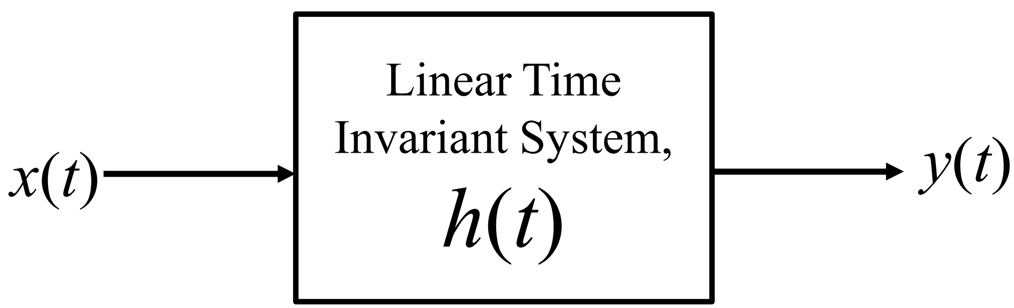

We will find relation betn $x(t)$ and $y(t)$